



DEPARTMENT OF
**COMPUTER
SCIENCE**

Natural Disaster Prediction Model on Earthquakes (PME)

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Table of Contents

- Introduction
- Problem Statement
- Contribution
- Methodology
- Goals & Objectives
- Application Architecture
- Tools & Technologies



Natural Disaster Prediction Model on Earthquakes



1

Introduction

Let's start right away!

How many people die from earthquakes every year?



<https://www.statista.com/statistics/263108/global-death-toll-due-to-earthquakes-since-2000/>

10,000

On average, a magnitude 8 earthquake strikes somewhere every year, and some 10,000 people die in earthquakes annually

Lets go

back in history
first !



“ On 8th October 2005
Kashmir earthquake
occured which registered a
moment magnitude of 7.6
and had a maximum
Mercalli intensity of VIII

100,000+

People died.

138,000

Injured

3.5 Million

Rendered homeless.



Big Motivation

- Now we will be able to predict earthquakes of different magnitudes as a result now there will be less false alarms which means many lives saved and less financial losses.

Maps



Introduction

- Earthquakes are one of the major catastrophe and their unpredictability causes even more destruction in terms of human life and financial losses.
- The seismologist community has been unsuccessful in developing methods to predict earthquakes.

Introduction

□ Lack of technology in accurately monitoring stress changes, pressure and temperature variations deep beneath the crust. Unavailability of comprehensive data about seismic features.

□ Gap between seismologists and computer scientist.

2

Problem Statement

What was the main problem?

|The **main problem** is to develop a model that can predict **earthquakes** using data on **seismic features** and give accurate results, such as the previous models were not accurate enough in terms of prediction of earthquake on different **magnitudes**.

Consequences of a false prediction?

- ❑ Economically costly
- ❑ Undermine confidence
- ❑ Effectiveness of warnings

3

Goals & Objectives

Our mission?

Goals & Objectives



Goals



Make future earthquake prediction



Make alarming condition to save lives and financial loss.



Make efficient way to predict earthquake.



Objectives



To test the training model with data.



To compare the model with existing techniques.

What constitutes a valid prediction?

- Date and Time
- Place of occurrence
- Magnitude
- Recorded in written form
- Confidence estimate (Probability)



4

Contribution

How everything supported us?

Contribution

- ❑ Combination of various machine learning techniques.
- ❑ Dataset divided into training & testing data.
- ❑ 70% for training.
- ❑ 30% for testing.
- ❑ Separate results for every region.
- ❑ Configurations and setups arranged same.
- ❑ Every region has different properties and classified

5

Methodology

What steps we took to reach our goal?

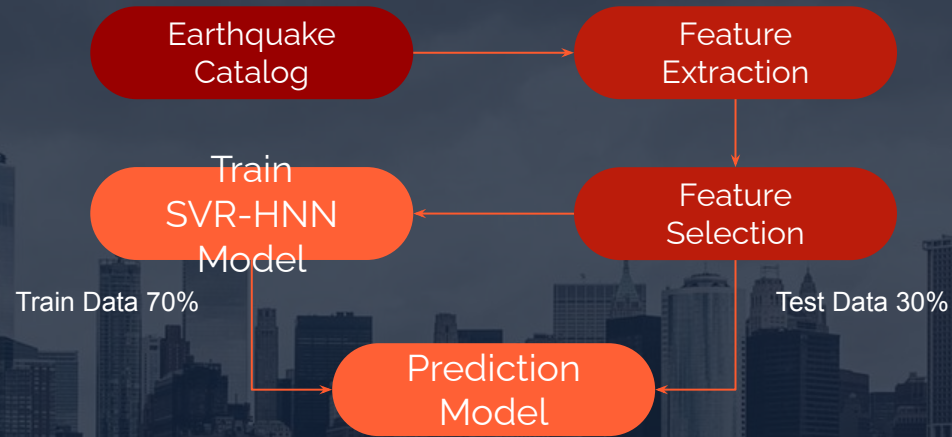
Dataset

- We are in collaboration with NCP (National Center for Physics).
- We will use their provided dataset.
- Dataset will be in the form of excel sheets.

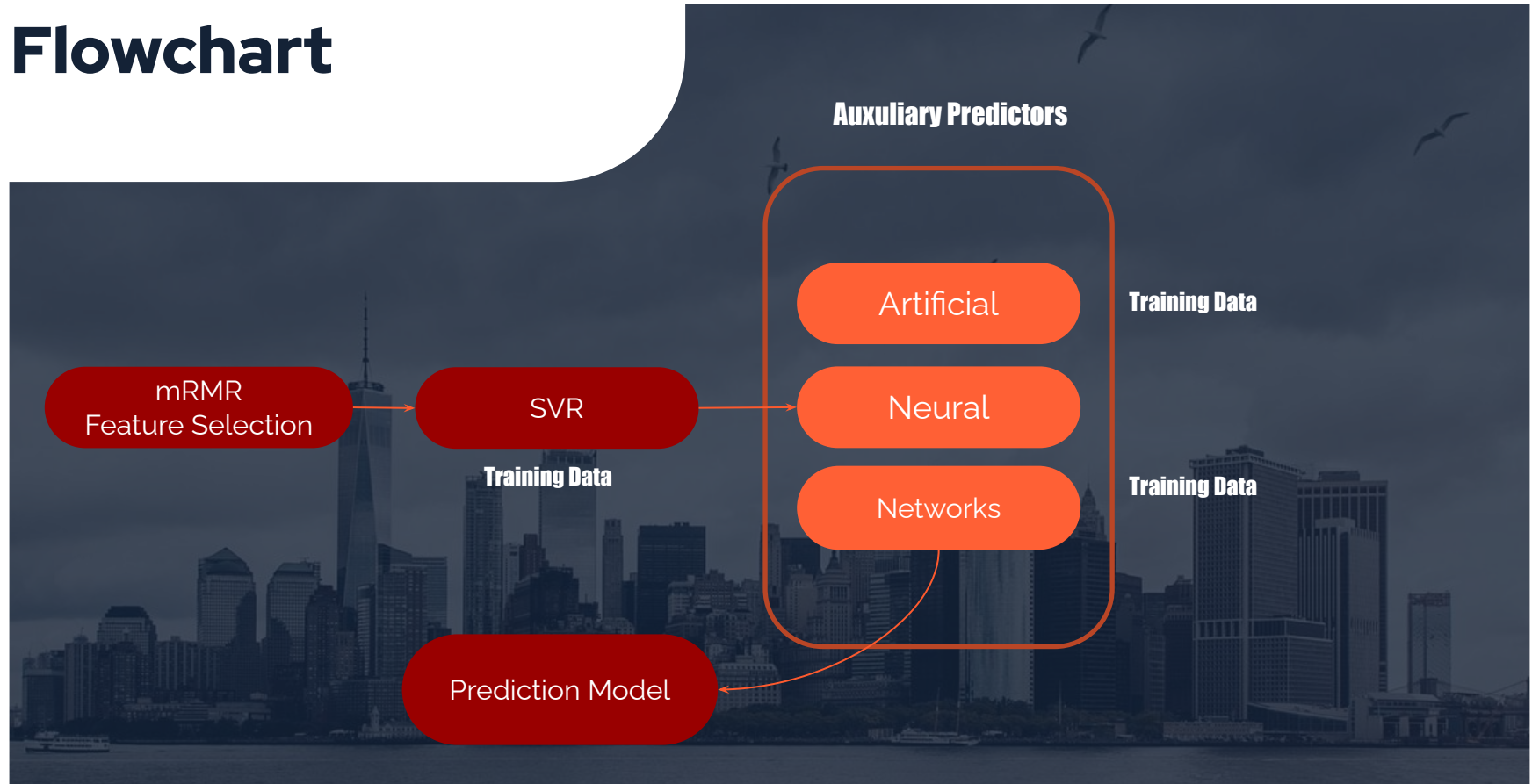
Our **process** is easy



Flowchart for Research Methodology

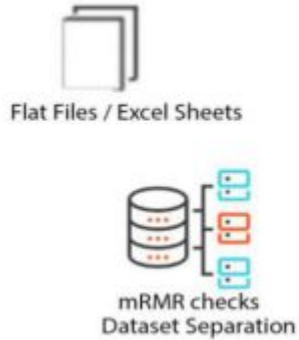


Flowchart

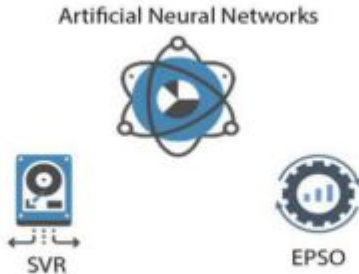


Application Architecture

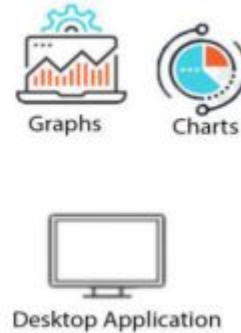
Database Layer



Application Layer



Presentation Layer



Tools & Technologies

❖ Programming Language

- Matlab
- Python
- C#

❖ Environment

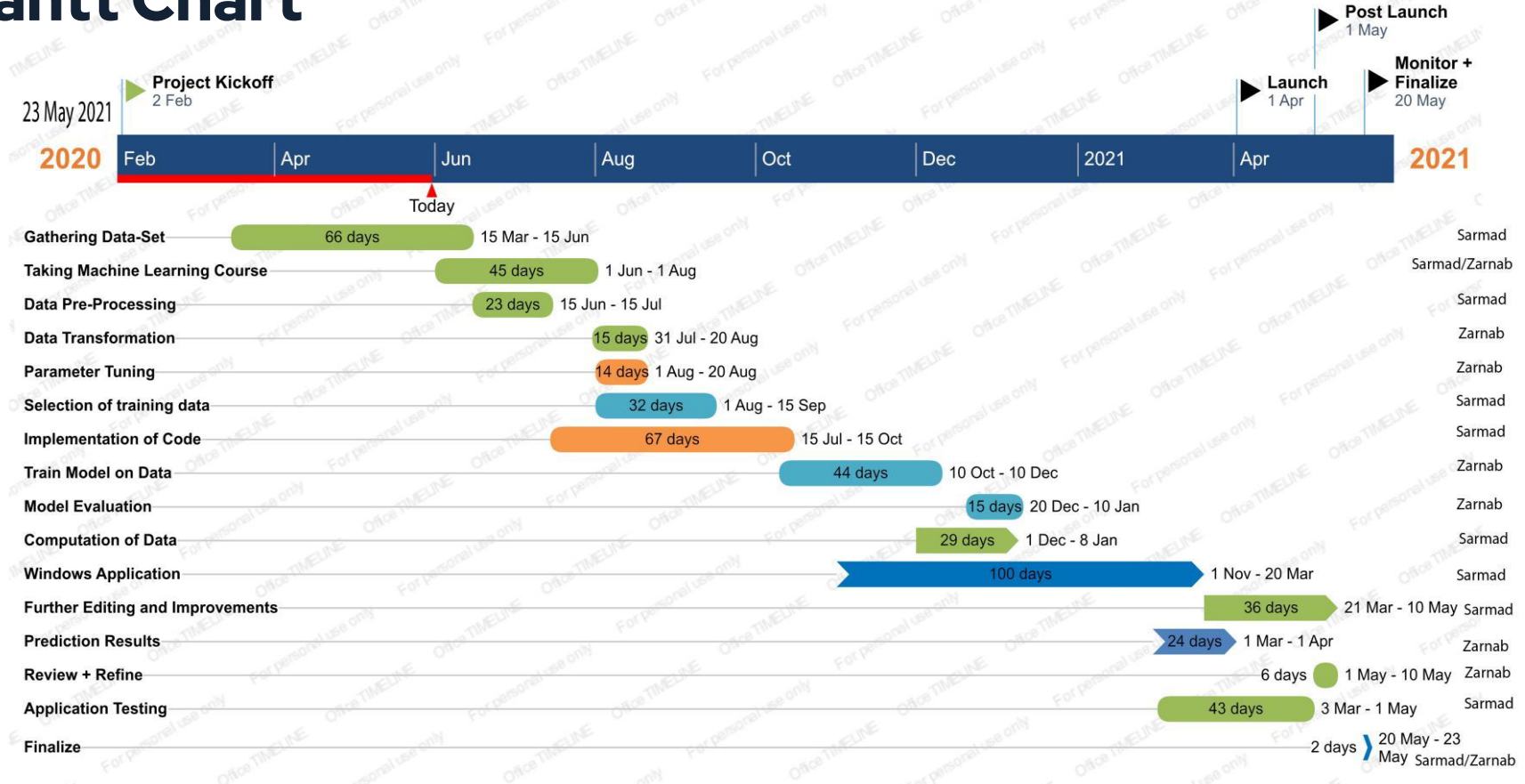
- Visual Studio
- PyCharm
- Matlab Application Software

Desktop project

Our project also includes a desktop application which will display the results of our prediction model graphically.



Gantt Chart



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Thanks!

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